

Dual 18A or Single 36A DC/DC μ Module Regulator

FEATURES

- Dual 18A or Single 36A Output
- Wide Input Voltage Range: 4.5V to 15V
- Output Voltage Range: 0.6V to 1.8V
- $\pm 1.5\%$ Maximum Total DC Output Error Over Line, Load and Temperature
- Differential Remote Sense Amplifier
- Current Mode Control/Fast Transient Response
- Adjustable Switching Frequency
- Overcurrent Foldback Protection
- Multiphase Parallel Current Sharing with Multiple LTM4630s Up to 144A
- Frequency Synchronization
- Internal Temperature Monitor
- Pin Compatible with the LTM4620 and LTM4620A (Dual 13A, Single 26A) and LTM4628 (Dual 8A, Single 16A)
- Selectable Burst Mode[®] Operation
- Soft-Start/Voltage Tracking
- Output Overvoltage Protection
- 16mm $\times$ 16mm $\times$ 4.41mm LGA and 16mm $\times$ 16mm $\times$ 5.01mm BGA Packages

APPLICATIONS

- Telecom and Networking Equipment
- Storage and ATCA Cards
- Industrial Equipment

DESCRIPTION

The **LTM4630** is a dual 18A or single 36A output switching mode step-down DC/DC μ Module[®] (micromodule) regulator. Included in the package are the switching controllers, power FETs, inductors, and all supporting components. Operating from an input voltage range of 4.5V to 15V, the LTM4630 supports two outputs each with an output voltage range of 0.6V to 1.8V, each set by a single external resistor. Its high efficiency design delivers up to 18A continuous current for each output. Only a few input and output capacitors are needed. The LTM4630 is pin compatible with the LTM4620 and LTM4620A (dual 13A, single 26A) and the LTM4628 (dual 8A, single 16A).

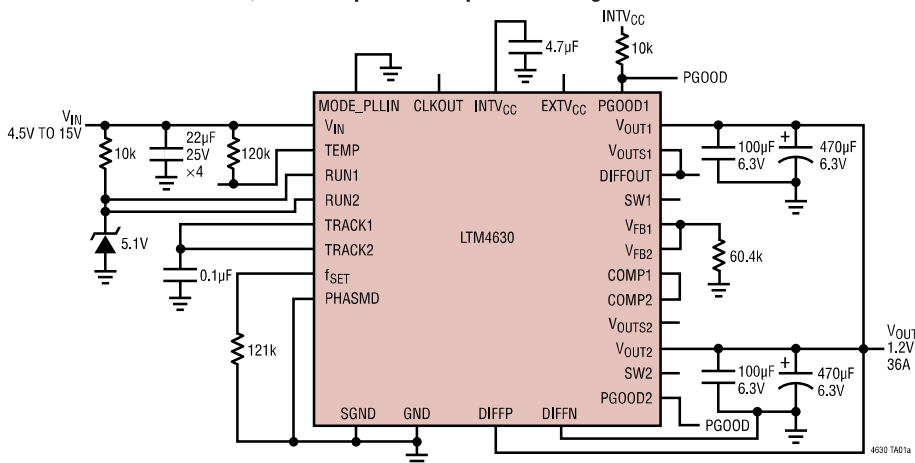
The device supports frequency synchronization, multi-phase operation, Burst Mode operation and output voltage tracking for supply rail sequencing and has an onboard temperature diode for device temperature monitoring. High switching frequency and a current mode architecture enable a very fast transient response to line and load changes without sacrificing stability.

Fault protection features include overvoltage and overcurrent protection. The LTM4630 is offered in 16mm $\times$ 16mm $\times$ 4.41mm LGA and 16mm $\times$ 16mm $\times$ 5.01mm BGA packages. The LTM4630 is ROHS compliant.

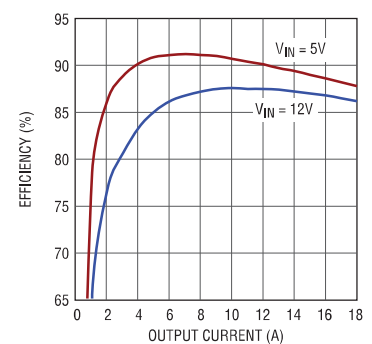
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TYPICAL APPLICATION

36A, 1.2V Output DC/DC μ Module Regulator



1.2V_{OUT} Efficiency vs I_{OUT}



ELECTRICAL CHARACTERISTICS The ● denotes the specifications which apply over the specified internal operating temperature range. Specified as each individual output channel. $T_A = 25^\circ\text{C}$ (Note 2), $V_{IN} = 12\text{V}$ and V_{RUN1} , V_{RUN2} at 5V unless otherwise noted. Per the typical application in Figure 23.

SYMBOL	PARAMETER	CONDITIONS		MIN	TYP	MAX	UNITS
V _{IN}	Input DC Voltage		●	4.5		15	V
V _{OUT}	Output Voltage		●	0.6		1.8	V
V _{OUT1(DC)} , V _{OUT2(DC)}	Output Voltage, Total Variation with Line and Load	C _{IN} = 22μF × 3, C _{OUT} = 100μF × 1 Ceramic, 470μF POSCAP V _{IN} = 12V, V _{OUT} = 1.5V, I _{OUT} = 0A to 18A	●	1.477	1.5	1.523	V

Input Specifications

V_{RUN1} , V_{RUN2}	RUN Pin On/Off Threshold	RUN Rising		1.1	1.25	1.40	V
$V_{RUN1HYS}$, $V_{RUN2HYS}$	RUN Pin On Hysteresis				150		mV
$I_{INRUSH(VIN)}$	Input Inrush Current at Start-Up	$I_{OUT} = 0\text{A}$, $C_{IN} = 22\mu\text{F} \times 3$, $C_{SS} = 0.01\mu\text{F}$, $C_{OUT} = 100\mu\text{F} \times 3$, $V_{OUT1} = 1.5\text{V}$, $V_{OUT2} = 1.5\text{V}$, $V_{IN} = 12\text{V}$			1		A
$I_{Q(VIN)}$	Input Supply Bias Current	$V_{IN} = 12\text{V}$, $V_{OUT} = 1.5\text{V}$, Burst Mode Operation $V_{IN} = 12\text{V}$, $V_{OUT} = 1.5\text{V}$, Pulse-Skipping Mode $V_{IN} = 12\text{V}$, $V_{OUT} = 1.5\text{V}$, Switching Continuous Shutdown, $RUN = 0$, $V_{IN} = 12\text{V}$			3 15 65 50		 mA mA mA μA
$I_{S(VIN)}$	Input Supply Current	$V_{IN} = 5\text{V}$, $V_{OUT} = 1.5\text{V}$, $I_{OUT} = 18\text{A}$ $V_{IN} = 12\text{V}$, $V_{OUT} = 1.5\text{V}$, $I_{OUT} = 18\text{A}$			6 2.6		 A A

Output Specifications

$I_{OUT1(DC)}$, $I_{OUT2(DC)}$	Output Continuous Current Range	$V_{IN} = 12\text{V}$, $V_{OUT} = 1.5\text{V}$ (Note 7)		0		18	A
$\Delta V_{OUT1(LINE)}/V_{OUT1}$ $\Delta V_{OUT2(LINE)}/V_{OUT2}$	Line Regulation Accuracy	$V_{OUT} = 1.5\text{V}$, V_{IN} from 4.5V to 15V $I_{OUT} = 0\text{A}$ for Each Output,	●		0.01	0.025	%/V
$\Delta V_{OUT1}/V_{OUT1}$ $\Delta V_{OUT2}/V_{OUT2}$	Load Regulation Accuracy	For Each Output, $V_{OUT} = 1.5\text{V}$, 0A to 18A $V_{IN} = 12\text{V}$ (Note 7)	●		0.5	0.75	%
$V_{OUT1(AC)}$, $V_{OUT2(AC)}$	Output Ripple Voltage	For Each Output, $I_{OUT} = 0\text{A}$, $C_{OUT} = 100\mu\text{F} \times 3$ / X7R/Ceramic, 470 μF POSCAP, $V_{IN} = 12\text{V}$, $V_{OUT} = 1.5\text{V}$, Frequency = 450kHz			15		mV _{p-p}
f_S (Each Channel)	Output Ripple Voltage Frequency	$V_{IN} = 12\text{V}$, $V_{OUT} = 1.5\text{V}$, $f_{SET} = 1.25\text{V}$ (Note 4)			500		kHz
f_{SYNC} (Each Channel)	SYNC Capture Range			400		780	kHz
$\Delta V_{OUTSTART}$ (Each Channel)	Turn-On Overshoot	$C_{OUT} = 100\mu\text{F}/\text{X5R}/\text{Ceramic}$, 470 μF POSCAP, $V_{OUT} = 1.5\text{V}$, $I_{OUT} = 0\text{A}$ $V_{IN} = 12\text{V}$			10		mV
t_{START} (Each Channel)	Turn-On Time	$C_{OUT} = 100\mu\text{F}/\text{X5R}/\text{Ceramic}$, 470 μF POSCAP, No Load, TRACK/SS with 0.01 μF to GND, $V_{IN} = 12\text{V}$			5		ms
$\Delta V_{OUT(LS)}$ (Each Channel)	Peak Deviation for Dynamic Load	Load: 0% to 50% to 0% of Full Load $C_{OUT} = 22\mu\text{F} \times 3/\text{X5R}/\text{Ceramic}$, 470 μF POSCAP $V_{IN} = 12\text{V}$, $V_{OUT} = 1.5\text{V}$			30		mV
t_{SETTLE} (Each Channel)	Settling Time for Dynamic Load Step	Load: 0% to 50% to 0% of Full Load, $V_{IN} = 12\text{V}$, $C_{OUT} = 100\mu\text{F}$, 470 μF POSCAP			20		μs
$I_{OUT(PK)}$ (Each Channel)	Output Current Limit	$V_{IN} = 12\text{V}$, $V_{OUT} = 1.5\text{V}$			30		A

Control Section

V_{FB1} , V_{FB2}	Voltage at V_{FB} Pins	$I_{OUT} = 0\text{A}$, $V_{OUT} = 1.5\text{V}$	●	0.592	0.600	0.606	V
I_{FB}		(Note 6)			-5	-20	nA
V_{OVL}	Feedback Overvoltage Lockout		●	0.64	0.66	0.68	V
TRACK1 (I), TRACK2 (I)	Track Pin Soft-Start Pull-Up Current	TRACK1 (I), TRACK2 (I) Start at 0V		1	1.25	1.5	μA

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SYMBOL	PARAMETER	CONDITIONS	MIN	TYP	MAX	UNITS
UVLO	Undervoltage Lockout (Falling)			3.3		V
UVLO Hysteresis				0.6		V
$t_{ON(MIN)}$	Minimum On-Time	(Note 6)		90		ns
R_{FBH1} , R_{FBH2}	Resistor Between V_{OUTS1} , V_{OUTS2} and V_{FB1} , V_{FB2} Pins for Each Output		60.05	60.4	60.75	$k\Omega$
V_{PGOOD1} , V_{PGOOD2} Low	PGOOD Voltage Low	$I_{PGOOD} = 2\text{mA}$		0.1	0.3	V
I_{PGOOD}	PGOOD Leakage Current	$V_{PGOOD} = 5\text{V}$			± 5	μA
V_{PGOOD}	PGOOD Trip Level	V_{FB} with Respect to Set Output Voltage V_{FB} Ramping Negative V_{FB} Ramping Positive		-10 10		% %

INTV_{CC} Linear Regulator

V_{INTVCC}	Internal V_{CC} Voltage	$6\text{V} < V_{IN} < 15\text{V}$	4.8	5	5.2	V
V_{INTVCC} Load Regulation	INTV _{CC} Load Regulation	$I_{CC} = 0\text{mA}$ to 50mA		0.5	2	%
V_{EXTVCC}	EXTV _{CC} Switchover Voltage	EXTV _{CC} Ramping Positive	4.5	4.7		V
$V_{EXTVCC(DROP)}$	EXTV _{CC} Dropout	$I_{CC} = 20\text{mA}$, $V_{EXTVCC} = 5\text{V}$		50	100	mV
$V_{EXTVCC(HYST)}$	EXTV _{CC} Hysteresis			220		mV

Oscillator and Phase-Locked Loop

Frequency Nominal	Nominal Frequency	$f_{SET} = 1.2\text{V}$	450	500	550	kHz
Frequency Low	Lowest Frequency	$f_{SET} = 0\text{V}$ (Note 5)	210	250	290	kHz
Frequency High	Highest Frequency	$f_{SET} > 2.4\text{V}$, Up to INTV _{CC}	700	780	860	kHz
f_{SET}	Frequency Set Current		9	10	11	μA
R_{MODE_PLLIN}	MODE_PLLIN Input Resistance			250		$k\Omega$
CLKOUT	Phase (Relative to V_{OUT1})	PHASMD = GND PHASMD = Float PHASMD = INTV _{CC}		60 90 120		Deg Deg Deg
CLK High CLK Low	Clock High Output Voltage Clock Low Output Voltage		2		0.2	V V

Differential Amplifier

A_V Differential Amplifier	Gain			1		V/V
R_{IN}	Input Resistance	Measured at DIFFP Input		80		$k\Omega$
V_{OS}	Input Offset Voltage	$V_{DIFFP} = V_{DIFFOUT} = 1.5\text{V}$, $I_{DIFFOUT} = 100\mu\text{A}$			3	mV
PSRR Differential Amplifier	Power Supply Rejection Ratio	$5\text{V} < V_{IN} < 15\text{V}$		90		dB
I_{CL}	Maximum Output Current			3		mA
$V_{OUT(MAX)}$	Maximum Output Voltage	$I_{DIFFOUT} = 300\mu\text{A}$	INTV _{CC} - 1.4			V
GBW	Gain Bandwidth Product			3		MHz
V_{TEMP}	Diode Connected PNP	$I = 100\mu\text{A}$		0.6		V
TC	Temperature Coefficient	●		-2.2		mV/C